Beyond Personality and Fingerprints: Behavioral Evidence for Pedagogical Policy Signatures in AI Tutors

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Abstract

Questionnaire studies of language-model personality are vulnerable to prompt fragility and a self-report–behavior gap. We instead study *pedagogical policy signatures*: recurring, model-conditioned teaching actions in authentic educational responses. Across 31,638 paired responses from six models, transparent policy features identify the model on held-out teaching families (0.242 accuracy versus 0.167 chance), but generic tasks yield stronger attribution. In a nine-model, 26,586-response panel, the same pedagogy instruction makes every model ask more questions and produce shorter turns, while reducing cross-model policy dispersion to 0.763 of baseline (95% CI [0.738, 0.794]). A classifier trained on 18,541 human-labelled teacher turns finds higher average next-action agreement after prompting, but agreement collapses when the human teacher selected direct telling. Transparent policy features improve leave-one-model-out tutoring-quality AUC by 0.023–0.031 beyond exact-item controls, while judged adaptive teaching does not improve human-gold learner diagnosis. In a prospectively frozen 2,560-response factorial, question and answer clauses change validated target behaviors by 0.773 and 0.895; a warm-tone clause changes a frozen encouragement marker by 0.595. An independently ordered 640-response replication preserves all three operational effects, whereas learner requests fail both adaptation gates. A blind three-judge audit validates the question and answer detectors but rejects semantic warmth for the marker. Finally, 1,074 complete blinded judge–batch annotations retain help directness and cognitive load as cross-task signatures, but only help directness as a validated disposition. The evidence supports stable and controllable policy signatures in this finite panel, not human-like personality or learning benefit.

1 Introduction

Language models are often described as having personalities. Much evidence comes from asking them to complete human inventories or enact a persona (Jiang et al., 2024; Lee et al., 2025). Such studies establish expressibility, but not whether an elicited profile predicts behavior when a model tutors a learner. Wording and option order destabilize scores (Shu et al., 2024); recent work also questions construct equivalence and predictive validity (Han et al., 2025; Zierahn et al., 2026; Contreras, 2026).

Educational agents offer a consequential behavioral test. A tutor must decide whether to explain or elicit, reveal an answer or preserve learner effort, and diagnose a misconception or remain cautious. These actions are observable across repeated contexts and are not universally good: a Socratic question can help after a productive error and obstruct when direct instruction is warranted.

We introduce a behavior-first hierarchy. A *surface fingerprint* is any stable authorship cue. A *pedagogical policy signature* is an interpretable teaching-action pattern surviving exact-item and surface controls. A *validated pedagogical disposition* additionally recurs across educational tasks, relates to an independent criterion, and responds coherently to an intervention. These terms do not imply human traits or inner mental states.

Our paired archive contains the same benchmark items and harness across model systems. Held-out-task analyses test recurrence; generic tasks expose domain-general fingerprints; paired generic and explicit-pedagogy prompts supply a within-model intervention; and existing human teacher moves and human-gold learner-state tasks provide criteria independent of the primary quality judge. The results reject both a verbosity-only account and a universal beneficial persona. They support a narrower, actionable object: finite-panel default tutoring policies and their resistance to steering.

The paper contributes four linked tests rather than a new personality inventory: (1) same-item cross-task signatures with generic-domain controls; (2) a paired prompt intervention measuring both convergence and action-specific harm; (3) independent criteria from human teacher actions and human-gold learner diagnosis; and (4) a blinded semantic panel whose frozen cumulative rule can downgrade, but not selectively upgrade, the headline construct.

2 Related Work

Behavior-first psychometrics is an active direction. GenPT generates projective stimuli rather than fixed inventories (Wang et al., 2026), while construct-matched contexts can improve self-report–behavior coherence (Kocielnik et al., 2026). Our unit is narrower and more ecological: archived tutor actions on shared tasks, subjected to task, fingerprint, intervention, and external-criterion controls.

MathDial defines teacher moves and premature solution reveal (Macina et al., 2023); MRBench and MathTutorBench organize open-ended tutoring capabilities (Maurya et al., 2025; Macina et al., 2025). LongTutor separates historical evidence, knowledge-state diagnosis, and adaptive teaching (Li et al., 2026). Pedagogical Alignment and StratL train or optimize a predefined scaffolded policy (Sonkar et al., 2024; Puech et al., 2025); prompt moderation can also harm relative to an unmoderated tutor (Steindl et al., 2025). Closest to our context-sensitivity question, Borchers and Shou (2025) find limited adaptivity under learner-context ablations. Tutor-persona steering (Lee et al., 2026) and theory-grounded runtime control (Nkambou et al., 2026) ask how a desired policy can be induced, while PATS adapts strategies to student personality (Rooein et al., 2026). We instead measure recurring deployed-system defaults, audit their construct validity, and then apply the same intervention. Work on tutor-strategy prediction (Ikram et al., 2025) and feedback failure localization (Yasir et al., 2026) further motivates reporting action-specific errors instead of one aggregate tutor score.

Answer withholding is already studied under pedagogical jailbreaks and multi-turn adversarial student pressure (Zhao et al., 2026b,a). Real deployment data also show that students commonly seek answers and that context can outweigh intended system design (Kobler et al., 2026). Our learner-action analysis is therefore not a first leakage or steering claim, and policy alignment is not treated as learning benefit.

More generally, IHEval finds sharp degradation when instruction priorities conflict (Zhang et al., 2025). Whether an explicit learner request survives system-level policy clauses is therefore an instruction-hierarchy test, not evidence of empathy, learner understanding, or pedagogical appropriateness.

Attribution alone is insufficient because lexical features identify generators across domains (McGovern et al., 2025). Prompted students are also not assumed to measure learning: recent evaluations find weak linguistic, behavioral, and cognitive fidelity (Scarlatos et al., 2026). We therefore treat objective diagnosis and evidence retrieval as prerequisite competence and reserve learning gain for prospective work.

3 Data and Governance

The core panel contains six frozen deployed configurations: MiniMax-M3, MiniMax-M2.7, GLM-5.2, DeepSeek-V4-Pro, Doubao-Seed-2.0-Pro, and Qwen3.5-4B. We retain only successful responses available for every model on an item. Six teaching arms yield 31,638 responses; four generic negative-control families yield 57,516. A nine-model MathDial panel adds three configurations for 26,586 responses. LongTutor contributes long learner histories, and MathTutorBench provides paired generic and LearnLM-style pedagogy prompts in standard and hard contexts.

The MathDial source supplies human teacher moves (`probing`, `focus`, `telling`, and `generic`). LongTutor supplies human-gold knowledge-state diagnosis and reference-scored historical evidence. We pin 84 relevant prediction/summary pairs by SHA-256. Exact provider-response regeneration is impossible: temperature is present for 22/84 runs, while seed and prompt-version metadata are absent from all 84.

Public analysis artifacts store identifiers, hashes, features, scores, and inferred labels rather than source responses. Although the LongTutor paper states CC BY 4.0 for data, we withhold longitudinal learner histories as a privacy-minimizing choice. The external semantic sample was transmitted only after explicit authorization.

Panel or criterion	Models	Shared units	Responses	Primary role
Core teaching	6	5,273 items	31,638	transparent signature
Generic controls	6	9,586 items	57,516	fingerprint falsifier
MathDial intervention	9	2,954 item–arms	26,586	steering and dispersion
Blinded semantics	6	358 context–arms	2,148	reliability and transport
LongTutor objective	6	1,000 histories	6,000	diagnosis boundary
Human teacher actions	–	18,541 turns	18,541	classifier training
Prospective factorial	5	32 problems	2,560	component addressability
Order replication	5	8 fixed problems	640	sequence robustness

Table 1: Evidence sources and dependence-preserving units. Shared units are not multiplied by the number of models when uncertainty is estimated.

4 Methods

4.1 Transparent features and transport

We deterministically count length, structure, formatting, questions, address terms, praise, encouragement, hedges, imperatives, history references, answer reveal, explanation, and diagnosis markers. Features are centered across models within exact item and standardized within benchmark. Multinomial logistic regression predicts model identity while holding out an entire task family. The same protocol on generic tasks is a negative control: strong generic attribution prevents relabeling an authorship fingerprint as pedagogy.

For paired prompts we subtract the generic-arm feature from the pedagogy-arm feature for the same context and model. Bootstrap resampling preserves the six correlated responses within each context. We measure both directional model-by-model effects and cross-model profile dispersion.

4.2 Quality, actions, and learner state

Quality prediction holds out one candidate model at a time. Exact-item effects form the baseline; length and policy features are nested additions. Byte-identical responses scored by two judges provide a judge swap. A separate set of 482 expert positive/negative pairs, each shown in both orders, audits three judges.

Three TF–IDF linear-SVM variants are trained on 18,541 human teacher turns, with five-fold splits grouped by math problem. Generated responses are then compared with the human next move. LongTutor analyses join teaching, diagnosis, and evidence tasks by exact history and model. Held-out-model prediction includes exact-history effects before teaching scores are added.

4.3 Blinded semantic panel and frozen hierarchy

The deterministic sample initially contained 360 context–arm batches and 2,160 responses: 40 LongTutor, 80 standard MathDial pairs, 40 hard pairs, and 80 Socratic contexts. Three judges coded eight bipolar 1–5 dimensions: help directness, elicitation, autonomy support, affective warmth, diagnostic specificity, personalization, cognitive load, and epistemic caution. Model identity was hidden and response order independently randomized by judge and batch. Confirmatory scores are judge medians.

One DeepSeek annotation remained unavailable after three rounds of three attempts. Before formal analysis and without inspecting the affected scores, we excluded that generic batch and its prespecified pedagogy counterpart for all judges. The complete-case panel therefore contains 358 batches, 2,148 responses, and 1,074 annotations (0.56% technical attrition).

All eight dimensions remain reported. A frozen cumulative hierarchy requires (i) ICC(3,k)$\geq$ .60, adequate range, leave-one-judge profile $\rho \geq .70$, and no prompt sign reversal; (ii) nonzero exact-context model variance plus cross-task ICC or median task correlation $\geq$ .50; and (iii) an independent quality/action, prompt/action, or BH-corrected human-gold diagnosis criterion. Fewer than two validated dimensions forces the policy-signature thesis used here.

4.4 Prospective component intervention

Before inspecting any generated response or effect, we froze 32 synthetic elementary-math problems and a complete $2 \times 2 \times 2$ intervention over question-first, answer-reveal, and warm-tone system clauses. Each cell is crossed with learner requests for direct help or exploration and five reachable model routes (2,560 calls). Deterministic detectors score whether the first sentence-like segment is a question, whether the correct answer is revealed, and whether a frozen encouragement lexicon appears. Every cell, model, family, secondary surface outcome, and two- and three-way interaction remains reported.

A component must exceed its registered target-effect threshold (.60 for question and answer; .30 for tone), be positive in every model, and have a target effect at least twice its largest primary cross-effect. Learner-request adaptation has separate .10 thresholds. The parent queue was lexicographic, so, still blind to outcomes, we froze an eight-problem replay with byte-identical prompts, model-specific SHA-256 order, and exact balance of all 16 request–policy cells in every consecutive block. A claim is order-robust only if both panels pass and every model is positive in the matching parent subset and replication; the replay can only downgrade a claim.

After seeing aggregate effects, we froze a downgrade-only detector audit. An outcome-independent hash sample contains 480 responses (320 parent, 160 replay). Three blinded judges label all 48 ten-response batches without model, factor, detector, or effect information. Validation requires majority coverage $\geq .95$, balanced accuracy $\geq .90$ with bootstrap lower bound $\geq .80$, and Cohen's $\kappa \geq .70$; failure restricts reporting to the literal detector.

4.5 Inference, multiplicity, and freeze timing

The inferential unit follows the claim (Table 1). Cross-task attribution reports an unweighted macro-average over held-out families. Prompt intervals resample underlying contexts while retaining all model-linked rows; judge calls are first reduced to response medians and never counted as independent educational cases. Quality models report every held-out-model fold, and exact sign tests treat folds, not responses, as replicates. Prompt tests use BH correction over the 48 model $\times$ dimension contrasts within each difficulty; semantic diagnosis uses within-outcome BH correction.

The multi-judge plan was frozen at 311/1,080 annotations, all from the first judge and before any second- or third-judge result. Single-judge directions had been inspected, so this is a prospective multi-judge freeze, not a full preregistration. Manuscript decision thresholds were fixed at 637/1,080, before the third judge and all LongTutor annotations; a later multiplicity amendment also preceded all semantic LongTutor results. Analyzer hashes and the technical- attrition amendment are retained in the audit trail.

5 Results

5.1 Signatures and generic fingerprints

Teaching-task held-out attribution is 0.242 with policy features and 0.251 with all transparent features, versus 0.167 chance; length alone reaches 0.230. Several within-item-centered proxies recur across tasks, including mean sentence length (ICC 0.645), explanation (0.546), hedging (0.474), and response length (0.470). The negative control reverses the hoped-for ordering: generic combined features reach 0.372 and surface features 0.347. Cross-domain policy geometry is not preserved (Mantel $r = .296$, exact $p = .307$). Attribution thus detects stable generator signatures but is not construct validity by itself (Figure 1A).

5.2 A shared prompt compresses policies

All nine models ask more questions and produce shorter responses under the explicit pedagogy prompt in both difficulty sets. Prompted cross-model dispersion is 0.763 of generic baseline (95% bootstrap CI [0.738, 0.794]). Praise, diagnosis, imperatives, and answer reveal do not move uniformly, rejecting a single scalar "more pedagogical" interpretation (Figure 1B).

5.3 Quality prediction and judge audit

On 14,770 identical responses, the two quality judges show 0.808 exact agreement, 0.955 within half a point, Spearman 0.806, and quadratic $\kappa = .810$. An exact-item model reaches mean AUC 0.867. Length adds about 0.002; policy plus length reaches 0.898 under DeepSeek (+0.031) and 0.889 under MiniMax (+0.023), with every held-out-model fold improving. Exact one-sided sign tests are $p = .0312$ and $p = .0156$.

On 482 expert-labelled pairs, MiniMax-M3, GLM-5.2, and DeepSeek-V4-Pro agree with experts at 0.844, 0.839, and 0.817; position consistency is 0.900–0.919. The three-judge majority reaches 0.835 and does not beat the strongest judge, so it is not treated as human ground truth.

5.4 Human actions expose oversteering

Problem-grouped macro F1 is 0.559–0.571 across the action classifiers. All 54 classifier $\times$ difficulty $\times$ model contrasts improve overall action match, by 0.091–0.114 in standard and 0.145–0.180 in hard

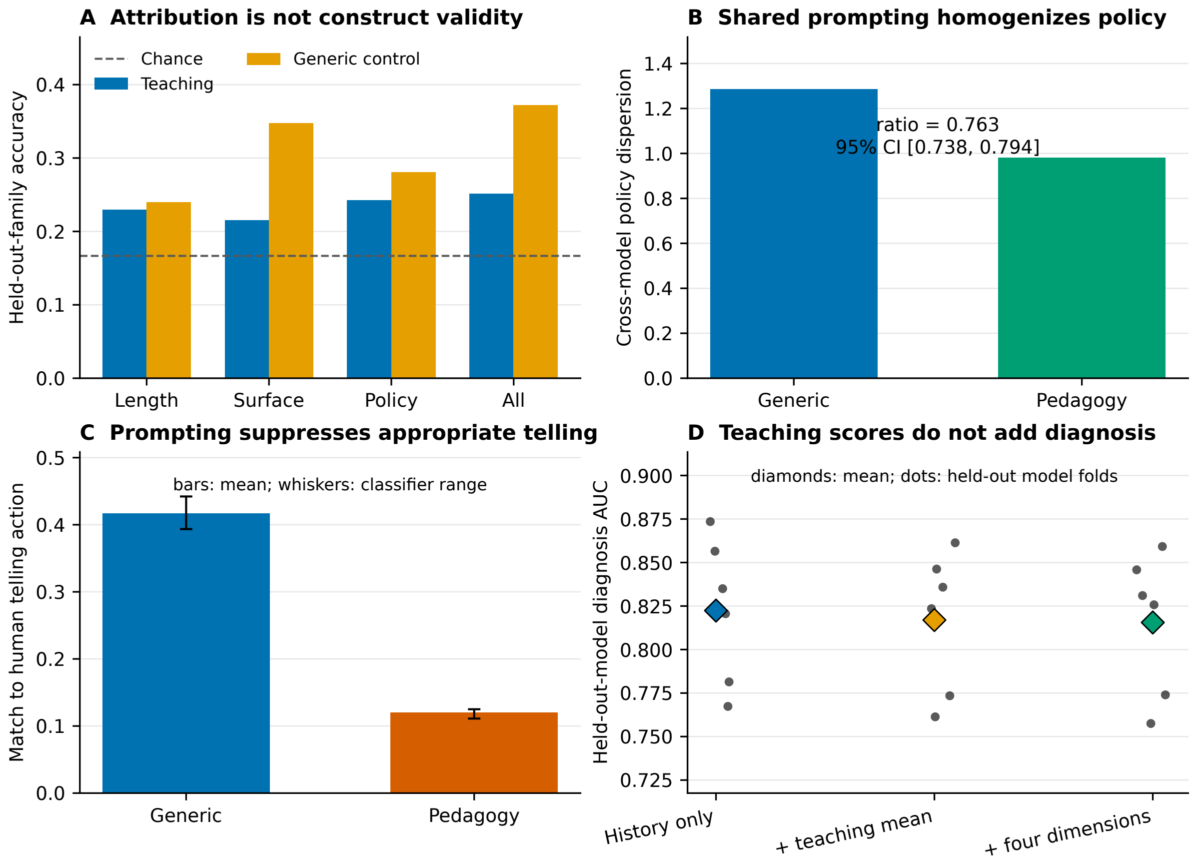

Figure 1: Complementary falsification tests. (A) Generic responses are more attributable than teaching responses. (B) a common pedagogy prompt compresses policy dispersion. (C) it reduces agreement when human teachers chose telling. (D) teaching dimensions do not improve human-gold diagnosis beyond exact history.

contexts. The hard subset is nearly all probing, so its gain is chiefly probing compliance.

The aggregate hides a consistent failure. For the 48 standard contexts where the human target is `telling`, generic match is 0.394–0.442 but prompted match falls to 0.111–0.125; inferred telling share drops 0.234–0.280. A universal question policy oversteers precisely where a human teacher chose direct instruction (Figure 1C).

5.5 Adaptive presentation is not diagnosis

Across 6,000 model–history observations, diagnosis accuracy ranges from 0.278 to 0.437. Teaching quality has weak within-model association with diagnosis (mean Spearman 0.173) and essentially none with evidence accuracy (-0.010). With exact-history effects and a held-out model, item-only diagnosis reaches AUC 0.822; four teaching dimensions reach 0.816 and their mean 0.817. Evidence RMSE remains 0.187 for all variants. Adaptive-sounding instruction does not substitute for learner-state inference (Figure 1D).

5.6 Confirmatory semantic measurement

Reliability is high for help directness (ICC(3,k)=0.905), elicitation (0.963), and cognitive load (0.855); epistemic caution fails the threshold (0.580). Dropping any judge preserves the full profile (ρ = .959–.962). Own-family judge residuals average only 0.001–0.014 points and candidate-position ranges are 0.075–0.160, below the registered 0.25 flags.

Semantic dimensions identify models on held-out tasks at 0.322, versus 0.167 chance, 0.239 for length, and 0.353 for transparent features. Combining semantic and transparent features reaches 0.391. Help directness (cross-task ICC 0.629; median task $\rho = .647$) and cognitive load (0.663; 0.677) qualify as signatures. Elicitation is reliable but misses stability (0.475; 0.468).

The intervention supplies a semantic manipulation check. The prompt reduces help directness by 1.709 points in standard and 1.542 in hard contexts, increases elicitation by 2.470 and 2.171, and reduces cognitive load by 0.759 and 0.721. Inde-

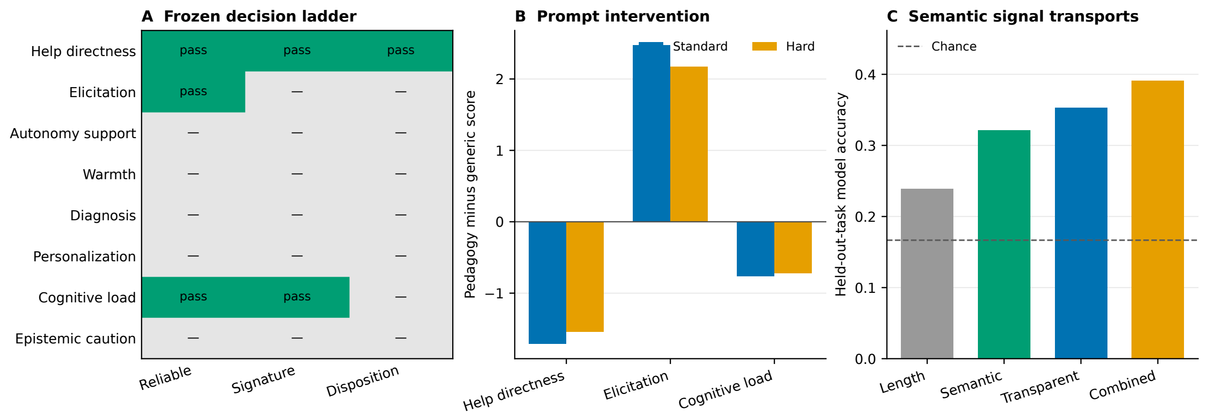

Figure 2: Confirmatory semantic results. (A) The frozen cumulative decision ladder retains two signatures and one validated disposition. (B) paired prompt effects. (C) held-out-task model attribution.

pendently inferred telling responses score 1.360 points higher in help directness than probing/focus. Help directness therefore passes prompt–action convergence. Semantic features add 0.015 mean quality AUC but fail fold consistency (minimum -0.004); no dimension survives BH correction against human-gold diagnosis. Consequently only help directness reaches the disposition tier, below the two-dimension threshold. Figure 2 shows how the frozen hierarchy prevents weaker dimensions from being promoted.

5.7 System clauses are addressable; requests are not

All three operational endpoints pass every registered parent gate and the downgrade-only order replication (Table 3). Effects are positive for all five models in the full panel, matching parent subset, and replication. By family, question effects span .750–.787 in the parent and .750–.863 in replication; answer effects span .878–.928 and .887–.975; encouragement-marker effects span .525–.659 and .487–.600. Thus no one model or problem family supplies the aggregate result.

The blind detector audit reaches balanced accuracy 1.000 (95% CI [1.000, 1.000], $\kappa = 1.000$) for question-first and .996 [.989, 1.000] ($\kappa = .992$) for correct-answer reveal. Both validate. The encouragement lexicon reaches only .818 [.787, .847] with $\kappa = .631$, failing two frozen gates. It therefore supports only a literal marker claim, not semantic warmth or support; this post-result audit cannot upgrade any claim.

Learner requests do not show corresponding robustness. Direct-help versus explore requests change correct reveal by only .073 [.061, .085] in the parent and .066 [.041, .087] in replication, below the registered .10 gate. Explore versus direct requests change question-first by $-.030$ [$-.047$, $-.014$] and $-.016$ [$-.037$, .006], respectively. This is instruction-hierarchy behavior, not reliable learner understanding.

Addressability also does not imply independence. The warm-tone clause reduces question-first by .118 in the parent and .103 in replication. The question$\times$tone interaction on question-first is $-.236$ [$-.266$, $-.208$] and $-.206$ [$-.256$, $-.162$], while the three-way question$\times$answer$\times$tone interaction is .203 [.138, .278] and .237 [.125, .350]. These prespecified interactions reproduce in sign and bound claims to controllable black-box components rather than orthogonal latent traits.

5.8 Robustness and negative evidence

The result is not driven by a single favorable judge. Full-profile leave-one-judge correlations are .959–.962; same-family judge residuals are .001–.014 points and position ranges .075–.160, below the registered .25 flags. Nor is a mean quality gain sufficient: semantic features add .015 mean AUC over transparent features but one held-out fold is $-.004$, so the quality gate fails. The 40-history semantic LongTutor subset also underperforms its exact-history diagnosis baseline (.744 versus .775), with no dimension surviving BH correction. These failures restrict Table 2 to one disposition.

6 Discussion

The results favor a narrower object than model personality. Tutor policies recur and are steerable, but some identity signal is a domain-general fingerprint. A shared intervention shows causal addressability

Dimension	ICC(3,k)	Task ICC	Median task ρ	Reversals	Highest retained tier
Help directness	.905	.629	.647	0	validated disposition
Elicitation	.963	.475	.468	0	reliable measurement
Autonomy support	.819	.610	.734	1	none
Affective warmth	.937	.479	.714	3	none
Diagnostic specificity	.924	.203	.371	1	none
Personalization	.832	.340	.257	1	none
Cognitive load	.855	.663	.677	0	cross-task signature
Epistemic caution	.580	.035	.129	0	none

Table 2: Every prespecified semantic dimension remains visible. Tiers are cumulative: reliability or stability alone cannot establish a disposition.

	Parent (2,560)			Replication (640)		
Clause	Effect [95% CI]	Selectivity	All models +	Effect [95% CI]	Selectivity	Robust
Question-first	.773 [.756, .790]	10.20	yes	.809 [.772, .841]	12.95	yes
Reveal answer	.895 [.883, .908]	11.35	yes	.922 [.900, .950]	9.52	yes
Encouragement lexicon	.595 [.563, .623]	5.04	yes	.550 [.497, .597]	5.33	yes

Table 3: Prospectively frozen component effects. Selectivity is the target effect divided by the largest absolute primary cross-effect; the registered threshold is 2.0. The final column also requires every model to be positive in the matched parent subset and independently ordered replication.

while also homogenizing models and suppressing appropriate telling. The learner-diagnosis null further separates adaptive presentation from accurate learner modeling. The prospective factorial sharpens the control result: explicit system clauses selectively address question-first and answer-reveal behavior, while the warm-tone clause changes only a non-semantic lexicon marker under the validated measurement boundary. Ordinary learner requests do not reliably induce matching adaptation. Stable cross-component interactions further argue for a controller over composable actions, not a menu of independent persona sliders.

This suggests a controller decomposition: estimate learner state, select an appropriate action, then realize that action in language. A model's signature describes its default and resistance to steering; it is not itself a quality label. Runtime grounding such as PedRAG may provide a controller layer, but our telling failure implies that the retrieval or steering policy must condition on learner state rather than enforce one pedagogy uniformly.

7 Reproducibility and Release

The release contains analysis code, frozen derived tables, figures, exclusion rules, prospective protocols, and a SHA-256 manifest, but no source responses or learner histories. The factorial release retains prompt/response hashes and deterministic metrics rather than provider text. A single command reruns the private-input semantic finalizer and blocks on coverage, claims, privacy structure, figures, syntax, or tests. Separately, a fresh checkout verified all frozen public paths and hashes. GitHub CI provisions Python 3.13 on a new Ubuntu runner, installs exact dependency versions, and repeats tests, registered claims, privacy auditing, figure rebuilding, and a clean-worktree check. This verifies the public package; it does not recreate private inputs or provider responses. The anonymous TeX build pins and hash-checks the official ACL style and produces a byte-stable PDF.

8 Conclusion

AI tutors in this finite panel exhibit stable and causally steerable pedagogical policy signatures. They are neither reducible to length nor equivalent to human personality. Their consequences are conditional: prompting can improve average action agreement while suppressing appropriate telling, and adaptive-sounding responses can coexist with weak learner diagnosis. Explicit system clauses can selectively control validated question-first and answer-reveal behaviors plus a literal encouragement marker under order replication, while learner requests alone do not reliably personalize them. Behavior-first measurement makes these distinctions auditable.

Limitations

The core panel contains only six systems. Thousands of responses increase precision for context effects but do not create model replicates; population and family claims are excluded. Most quality outcomes are LLM-judged, although judge swaps, expert-pair calibration, human action labels, and human-gold diagnosis provide complementary checks. Human action agreement is not learning gain, and no simulated-student claim is made. A learning-effectiveness claim requires prospective learners or a separately validated simulator and objective pre/post tests.

The prospective factorial uses synthetic elementary-math contexts and surface detectors. A post-result blind audit validates question-first and correct-answer reveal, but the encouragement lexicon fails semantic warmth validation and is reported literally. Its five endpoints are provider snapshots, and the eight-problem replay occurred later; provider-version drift and repeat-prompt caching cannot be separated from request-order robustness. Component effects are not learning outcomes, and reproduced interactions rule out treating the clauses as independent latent traits.

Provider system prompts, alias updates, and serving backends are uncontrolled. Generation cannot be exactly repeated because most settings and all seeds and prompt versions were not retained. Findings concern frozen deployed configurations, not immutable weights. The semantic complete-case rule was fixed after a provider failure but before formal analysis; its 0.56% paired technical attrition is fully reported.

Ethics Statement

The study analyzes existing model responses and educational histories. Public artifacts use hashes and derived measurements rather than raw learner histories or source responses. External semantic coding used only explicitly authorized endpoints and a frozen payload. Model-conditioned differences must not be read as human traits, demographic characteristics, or evidence that one system improves learning. Deployment should test learner outcomes and provide educator oversight, especially where a uniform Socratic policy may withhold needed direct help.

References

Conrad Borchers and Tianze Shou. 2025. Can large language models match tutoring system adaptivity? a benchmarking study. *arXiv preprint arXiv:2504.05570*.

Juan Manuel Contreras. 2026. An llm-native psychometric instrument reveals a self-report–behavior gap across 25 models. *arXiv preprint arXiv:2606.09843*.

Pengrui Han, Rafal Kocielnik, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, and R. Michael Alvarez. 2025. The personality illusion: Revealing dissociation between self-reports and behavior in llms. *arXiv preprint arXiv:2509.03730*.

Fareya Ikram, Alexander Scarlatos, and Andrew Lan. 2025. Exploring llms for predicting tutor strategy and student outcomes in dialogues. In *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications*, pages 765–779.

Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. 2024. PersonaLLM: Investigating the ability of large language models to express personality traits. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3605–3627.

Sebastian Kobler, Matthew Clemson, Angela Sun, and Jonathan K. Kummerfeld. 2026. Your students don't use llms like you wish they did. In *Proceedings of ACL 2026*, pages 19146–19170.

Rafal Kocielnik, Pengrui Han, Peiyang Song, Myrl G. Marmarelis, Ramit Debnath, Dean Mobbs, Anima Anandkumar, and R. Michael Alvarez. 2026. Rethinking psychometric evaluation of llms: When and why self-reports predict behavior. *arXiv preprint arXiv:2606.12730*.

Jaewook Lee, Alexander Scarlatos, Simon Woodhead, and Andrew Lan. 2026. Letting tutor personas speak up for llms: Learning steering vectors from dialogue via preference optimization. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications*, pages 78–92.

Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, and Youngjae Yu. 2025. Do llms have distinct and consistent personality? TRAIT: Personality testset designed for llms with psychometrics. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8412–8452.

Ning Li, Zheng Zhang, Zhenya Huang, Rui Li, Yi Zhan, Yinbo Luo, Qi Liu, and Enhong Chen. 2026. Long-Tutor: Benchmarking large language models for long-term personalized tutoring. In *Proceedings of ACL 2026*, pages 29712–29737.

Jakub Macina, Nico Daheim, Sakalan Chowdhury, Tanmay Sinha, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2023. MathDial: A dialogue tutoring dataset with rich pedagogical properties grounded in math reasoning problems. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5602–5621.

Jakub Macina, Nico Daheim, Ido Hakimi, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2025. MathTutorBench: A benchmark for measuring open-ended pedagogical capabilities of llm tutors. In *Proceedings of EMNLP 2025*, pages 204–221.

Kaushal Kumar Maurya, Kv Aditya Srivatsa, Kseniia Petukhova, and Ekaterina Kochmar. 2025. Unifying ai tutor evaluation: An evaluation taxonomy for pedagogical ability assessment of llm-powered ai tutors. In *Proceedings of NAACL-HLT 2025*, pages 1234–1251.

Hope McGovern, Rickard Stureborg, Yoshi Suhara, and Dimitris Alikaniotis. 2025. Your large language models are leaving fingerprints. In *Proceedings of the 1st Workshop on GenAI Content Detection*.

Roger Nkambou, Esperance Nfor, and Daril Kengne. 2026. PedRAG: Do llm tutors practice what they preach? preventing behavioral hallucinations through theory-grounded runtime control. In *Proceedings of the 19th International Conference on Educational Data Mining*. Poster and demo paper.

Romain Puech, Jakub Macina, Julia Chatain, Mrinmaya Sachan, and Manu Kapur. 2025. Towards the pedagogical steering of large language models for tutoring: A case study with modeling productive failure. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26291–26311.

Donya Rooein, Sankalan Pal Chowdhury, Mariia Eremeeva, Yuan Qin, Debora Nozza, Mrinmaya Sachan, and Dirk Hovy. 2026. PATS: Personality-aware teaching strategies with large language model tutors. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4186–4211.

Alexander Scarlatos, Jaewook Lee, Simon Woodhead, and Andrew Lan. 2026. Simulated students in tutoring dialogues: Substance or illusion? In *Proceedings of ACL 2026*, pages 42349–42385.

Bangzhao Shu, Lechen Zhang, Minje Choi, Lavinia Dunagan, Lajanugen Logeswaran, Moontae Lee, Dallas Card, and David Jurgens. 2024. You don't need a personality test to know these models are unreliable: Assessing the reliability of large language models on psychometric instruments. In *Proceedings of NAACL-HLT 2024*, pages 5263–5281.

Shashank Sonkar, Kangqi Ni, Sapana Chaudhary, and Richard Baraniuk. 2024. Pedagogical alignment of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 13641–13650.

Sebastian Steindl, Fabian Brunner, Nada Sissouno, Dominik Schwagerl, Florian Schöler-Niewiera, and Ulrich Schäfer. 2025. On the effectiveness of prompt-moderated llms for math tutoring at the tertiary level. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 11310–11323.

Ming Wang, Shuang Wu, Bixuan Wang, Lu Lin, Yuxin Chen, Xiaocui Yang, Daling Wang, Shi Feng, Yifei Zhang, and Yufan Sun. 2026. GenPT: Beyond self-report for reliable llm psychometrics via generative projective testing. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 40958–40974.

Tahreem Yasir, Wenbo Li, Sam Gilson, Sutapa Tithi, Xiaoyi Tian, and Tiffany Barnes. 2026. Confirming correct, missing the rest: Llm tutoring agents struggle where feedback matters most. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications*, pages 819–840.

Zhihan Zhang, Shiyang Li, Zixuan Zhang, Xin Liu, Haoming Jiang, Xianfeng Tang, Yifan Gao, Zheng Li, Haodong Wang, Zhaoxuan Tan, Yichuan Li, Qingyu Yin, Bing Yin, and Meng Jiang. 2025. IHEval: Evaluating language models on following the instruction hierarchy. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8374–8398.

Jin Zhao, Marta Knežević, and Tanja Käser. 2026a. Evaluating answer leakage robustness of llm tutors against adversarial student attacks. In *Proceedings of ACL 2026*, pages 30588–30617.

Sihang Zhao, Kangrui Yu, Youliang Yuan, Pinjia He, and Hongyi Wen. 2026b. SHAPE: Unifying safety, helpfulness and pedagogy for educational llms. In *Proceedings of ACL 2026*, pages 11537–11553.

Kim Zierahn, Cristina Cachero, Anna Korhonen, and Nuria Oliver. 2026. Llms aren't human: A critical perspective on llm personality. *arXiv preprint arXiv:2603.19030*.